When Free Lunch Is No Longer a Proxy for Student Poverty: The Community Eligibility Provision and State K–12 Funding*

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Abstract

I investigate how universal free school meals affect the measurement of student poverty and state K–12 funding. About half of U.S. public schools participate in the Community Eligibility Provision (CEP), which provides free meals to all students and eliminates household applications. This could distort free or reduced-price lunch (FRPL) data that many states use to allocate additional aid to low-income students. Using administrative data from 31 states, I find that reported FRPL shares diverge from child poverty in opposite directions depending on how states count students in CEP schools. Relative to poverty, FRPL shares fall by 5 log points in states reporting actual counts and rise by 10 log points in states counting all students in CEP schools. Despite these differences in reported FRPL, I find little evidence of funding loss. In the 15 states that allocate poverty funding based on FRPL, I can rule out annual losses in state formula aid exceeding $105 per student.

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1 Introduction

For decades, free and reduced-price lunch (FRPL) data from the Common Core of Data (CCD) have been the best available proxy for student poverty (National Research Council, 2012). Education researchers and policymakers have long used FRPL for accountability monitoring, program evaluation, and funding allocation. For example, more than 30 states used FRPL in their funding formulas to allocate additional aid to low-income students in 2019 (EdBuild, 2018).

However, the Community Eligibility Provision (CEP) has weakened FRPL as a measure of student poverty. Introduced nationwide in 2014, CEP allows high-poverty schools to provide free meals to all students and eliminates household applications, and roughly half of schools now participate (Hysom and FitzSimons, 2024). Without applications, schools can lose track of low-income students who do not participate in other public assistance programs. This could have unintended consequences for both FRPL data and state funding. Despite growing concern in policy circles over how CEP affects the measurement of student poverty (Chingos, 2016, 2018; Gutierrez and Blagg, 2026), little is known about how much CEP changes FRPL data or whether these changes reach state funding.

This paper sheds light on how CEP affects FRPL data quality and state funding to low-income students using linked administrative data on school meal applications, CCD FRPL data, and school district finances from 31 states. My main strategy uses a difference-in-differences (DiD) design comparing earlier-adopting districts with later-adopting ones within the same state with similar baseline poverty rates, exploiting the staggered timing of adoption. To benchmark the funding estimates, I also simulate aid under each state's formula.

I find that CEP reduces meal applications across states, but reported FRPL shares diverge from child poverty in opposite directions depending on how states report the data. CEP reduces the share of students approved by application by 14 percentage points, or 57 percent of the pre-adoption mean. In states that report actual FRPL counts, reported FRPL falls by 5 log points relative to Census child poverty. In states that count all or most students in CEP schools as FRPL-eligible, it rises by 10 log points. As CEP continues to expand, FRPL data increasingly reflect each state's reporting rules rather than student poverty.

Despite the measurement gap, I find limited pass-through to state general formula aid. The DiD point estimate is $7.4 per student per year, and it is not statistically significant. The 95 percent confidence interval rules out declines larger than about $105 per student, or 1.4 percent of general formula aid. The null is

robust to alternative comparison groups, samples, and estimators. The formula simulation implies a decline of $42 per student per year, and its confidence interval lies entirely within that of the DiD estimate.

Why does the decline in reported FRPL not translate into lost aid? First, the FRPL count falls much less than applications do, because many states expand direct certification (automatic eligibility through public assistance records) to use Medicaid and still collect alternative income forms to track low-income students. Second, states typically do not rely on FRPL alone to allocate poverty funding. They use direct certification, Census poverty estimates, or broader definitions of "disadvantage" or "at-risk" that also include English learners and foster youth (Fischer and Duncombe, 2024). Third, states adjusted their formulas to accommodate the loss of FRPL data. Massachusetts, for example, replaced FRPL with a direct-certification measure and raised its per-student rates so that districts would not lose funding (Massachusetts Department of Elementary and Secondary Education, 2015, 2017).

This paper's findings matter for both research and policy. For education researchers, I show that FRPL from the CCD is no longer comparable across states or over time. Depending on the state's reporting rule, CEP adoption can lower or increase reported FRPL relative to child poverty. Studies that use FRPL to control for poverty, to define low-income subgroups, or to measure achievement gaps may therefore mistake changes in measurement for changes in poverty, especially when they compare districts across states or before and after CEP adoption. For policymakers, my results suggest that monitoring changes in FRPL data and using alternative sources to identify low-income students remain important. States can expand free meals without large funding losses if they adapt how they measure student poverty.

This paper contributes to three strands of literature. First, it contributes to the literature on FRPL data as a measure of student poverty. FRPL has long been recognized as the best available, though imperfect, measure of student poverty (Michelmore and Dynarski, 2017; Domina et al., 2018): some eligible families do not participate (Gleason, 2008), and income reported on school meal applications is rarely verified (Harwell and LeBeau, 2010). Evidence on how CEP changes FRPL data comes largely from single states. In Missouri, every student in a CEP school is counted as FRPL-eligible. CEP therefore widens the gap between FRPL counts and the number of students whose family incomes qualify, and a high FRPL share at the school level no longer separates the poorest schools from better-off ones (Koedel and Parsons, 2021; Fazlul, Koedel and Parsons, 2023). Using Oregon records linked to tax data, Spiegel et al. (2025) find that direct certification measures school economic disadvantage more accurately than FRPL. To my knowledge, this paper provides the first multi-state causal estimates of how CEP changes the FRPL data that states report, and it shows that

the sign of the effect depends on how states report CEP students.

Second, it contributes to research on the effects of CEP and school meal programs, which has largely focused on student outcomes and yielded mixed findings. Cohen et al. (2021) review recent work across disciplines and document a range of positive associations between universal school meals and student outcomes. By contrast, Ayllón and Lado (2025), focusing on the economics literature, find that school meal programs in high-income countries have minimal causal effects on students' behavior, health, and educational outcomes after accounting for publication bias. Related to this paper, Rothbart, Schwartz and Gutierrez (2023) show that in New York State, increased federal reimbursements under CEP more than offset lost local food-service revenue, with no effect on instructional spending. This paper instead studies how CEP affects state K–12 aid across states and rules out large losses.

Third, this paper contributes to the literature on administrative burden in safety net programs. Learning, compliance, and psychological costs reduce take-up of public benefits (Herd and Moynihan, 2025), and removing them is a common policy goal. I show that removing burden in one program can relocate it to another. The school meal application served two purposes: it determined meal eligibility, and it produced the low-income counts that education agencies use to target resources. CEP removes the application for the first purpose but not the need for the second. As a result, districts in many states still distribute alternative income forms, which ask families for the same information without a meal benefit in return, and states build new data matches with SNAP and Medicaid records. The burden of identifying low-income students thus shifts from the meal program to schools, families, and state agencies. This paper highlights that policy reforms need a holistic approach, especially when they concern low-income children.

2 Background

2.1 The Community Eligibility Provision and Changes to FRPL Data

Traditionally, school meals are means-tested, targeting children with family income at or below 185 percent of poverty. To expand access to free meals and reduce administrative burden, the federal government introduced CEP. CEP allows high-poverty schools to offer free meals to all students. The program began as a pilot in selected states in 2011 and became available nationwide in 2014–15. Eligibility is based on the identified student percentage (ISP), which is the share of students certified for free meals because they receive other public assistance such as SNAP and Medicaid. A school, a group of schools, or an entire district

is eligible for CEP if its individual or pooled ISP is at least 40 percent. Participation decisions are made at the district level. The share of meals reimbursed at the highest (free-meal) rate equals 1.6 times the ISP, so schools with higher ISPs have larger incentives to adopt. Today, about half of public schools participate in CEP (Hysom and FitzSimons, 2024). Figure 1 shows the number of districts that first adopt CEP in each year from 2014–15 to 2024–25.

CEP schools no longer collect applications and cannot reliably track low-income students who do not participate in other public assistance programs. Prior to CEP, students were certified for subsidized meals in two ways. First, through direct certification, states match student records to enrollment in SNAP and other means-tested programs and certify matched students for free meals. Second, students who were not directly certified could submit an annual application. In 2013–14, about 25 million students, roughly half of public-school students, were certified for subsidized meals. About half were directly certified, and most of the rest were approved through applications.

CEP also changes how states report FRPL data to the CCD. Some states continue to report students income-eligible for FRPL only, based on direct certification and alternative income forms. Others report all students in CEP schools as FRPL-eligible, or report 1.6 times the ISP (Sinclair and Chen, 2020). Some states only report direct certification, and several states stopped reporting FRPL altogether.[1] As a result, by 2024–25, 17 percent of public-school students attended schools that did not report FRPL counts, and 7 percent attended CEP schools that reported FRPL based on the school's CEP status rather than individual eligibility (Figure 2). Nationally, the share of public-school students eligible for free or reduced-price meals rose from 54 percent in 2013 to 73 percent in 2024, while the share of children aged 6–17 below 185 percent of poverty fell from 40 to 31 percent (Appendix Figure A.1). Because all students in CEP schools are eligible, meal eligibility is no longer tied to family income.

2.2 State K–12 Funding and FRPL Data

To allocate K–12 funding, states typically set a base funding per student and then give additional weights in the funding formula for students with greater needs, such as students in special education, English language learners, and low-income students (Chingos and Blagg, 2017*b*). More than 30 states allocate additional funding to low-income students, and most of these states define them as students eligible for FRPL (EdBuild,

[1] In 2018–19, six jurisdictions reported only directly certified students (Delaware, the District of Columbia, Massachusetts, Oklahoma, Tennessee, and West Virginia), and Alaska and Ohio did so for a subset of schools (Sinclair and Chen, 2020).

2018; Appendix Figure A.2). For example, if a state gives each FRPL student an additional weight of 0.25 on an $8,000 base, a district with 60 percent FRPL students receives about $1,200 per student in extra aid. If CEP lowers the district's reported FRPL share by 2 percentage points, the district loses about $40 per student. Because K–12 finance is highly decentralized, funding for low-income students varies substantially across states.[2]

States differ in how they incorporate FRPL data in CEP schools for funding purposes (Appendix Figure A.6). Twenty states use alternative income forms to collect student income information. Seven states replace FRPL with direct certification as the poverty measure in both CEP and non-CEP schools. Six states multiply the ISP by a multiplier to approximate the share of low-income students. Five states use the most recent school meals data available, such as pre-CEP FRPL rates. Five states use Title I or Census poverty data, and the remaining eight do not allocate additional funding for poverty.

This state funding for low-income students matters for equity. Local funding is often regressive. Districts with mostly non-poor students can raise more through property taxes because their residents have more property wealth, while high-poverty districts can raise less. State funding tends to offset this gap. As a result, total funding is only modestly progressive: in 2013–14, districts attended by poor students received about 2.5 percent more per student than districts attended by non-poor students (Chingos, 2017; Chingos and Blagg, 2017*a*).

3 Data and Sample

3.1 Data

I combine six district-level data sources: CEP participation, school meal applications, reported FRPL counts from the CCD, two Census measures of child poverty, and school district finances. Unless otherwise noted, the analysis uses school years 2009–10 through 2019–20.

CEP Participation I obtain school-level CEP participation for school years 2013–14 through 2024–25 from the Food Research & Action Center (FRAC). FRAC compiles the lists of CEP schools from state departments of education. For each school, the data report enrollment, CEP participation status, and ISP. I aggregate school-level data to district level.

[2]Some states apply per-student weights, ranging from 5 percent of the base amount in Mississippi to 97 percent in Maryland, while others provide flat supplements, such as $1,818 per low-income student in New Hampshire in FY2018 (EdBuild, 2018).

School Meal Applications I obtain school meal application data for school years 2013–14 through 2019–20 through a public information request from the U.S. Department of Agriculture (USDA). The form FNS-742 reports, for each school district, the number of students approved for free or reduced-price meals by household application, the number directly certified, and total enrollment.

Common Core of Data (CCD) The CCD, maintained by the National Center for Education Statistics (NCES), collects annual data on all public schools and school districts. I aggregate school-level counts of students eligible for FRPL and total enrollment to the district level.

Small Area Income and Poverty Estimates (SAIPE) I use the Census Bureau's SAIPE estimates of the number of children aged 5–17 in families below the poverty line, which are available annually for every school district. I use them to benchmark need because they are more precise and timely than the 5-year ACS estimates. The U.S. Department of Education uses these estimates to allocate Title I funds. The estimates are model-based: county estimates combine direct ACS estimates with administrative records, such as federal tax returns and SNAP participation, and are then allocated to school districts.

American Community Survey (ACS) My alternative measure of need uses the 5-year ACS estimates of the share of children aged 6–17 in families with income below 185 percent of the poverty line, the income limit for reduced-price meals, for each school district (table B17024). I link each school year to the 5-year window ending in the fall of that year. For example, the 2011–2015 ACS is linked to the 2015–16 school year, so the analysis uses windows 2005–2009 to 2015–2019.

School District Finance Data I obtain district revenue from the F-33 School District Finance Survey, collected by the Census Bureau and NCES. My primary outcome is state general formula assistance per student (item C01), the main channel through which states allocate poverty funding. I also examine total state revenue, other state aid categories, federal and local child nutrition revenue, and federal Title I revenue. All monetary variables are adjusted to 2024 dollars using the CPI-U.

3.2 Key Variables

CEP adoption: A district is treated if at least one of its schools adopts CEP. The adoption year is the first year in which a school in the district participates, based on the FRAC data. This binary treatment captures

most of a district's exposure: in treated districts, over 70 percent of students are enrolled in CEP schools.

FRPL share: The FRPL share is the number of students eligible for free or reduced-price meals divided by enrollment in the district in the CCD.

Need: My primary need measure is the share of children aged 5–17 in families with income below 100 percent of the poverty line in SAIPE. As an alternative, I use the 5-year ACS share of children aged 6–17 with income below 185 percent of poverty.

FRPL share relative to need: I take the difference between the log FRPL share and the log SAIPE child poverty rate. Changes in this measure capture the percent change in reported FRPL relative to the percent change in poverty, so a decline means FRPL falls relative to poverty. Using logs means the difference in income thresholds (185 versus 100 percent of poverty) does not affect the estimates, assuming the ratio of near-poor children (income between 100 and 185 percent of poverty) to poor children within a district is stable over time.

State funding: My main funding outcome is state general formula assistance (F-33 item C01) per student. Most states build poverty weights into their main formula, so poverty funding is usually reported as general formula aid. Some states instead report it as compensatory aid (item C06), but only 12 states report meaningful amounts of compensatory aid, and states differ in how they split funding between the two items.[3] General formula aid is therefore the most comparable outcome across states.

3.3 Sample Construction

I construct a district-by-year panel for school years 2009–10 to 2019–20. I link the CCD and district finance data using the NCES district identifier. The FRAC and USDA application data do not include this identifier. I match FRAC to CCD using school and district name and state.[4] I match USDA application data to CCD using state, district name, and city.

For FRPL measurement, the sample includes 31 states that report FRPL data to the CCD consistently over the sample period. The first group includes 22 states that report actual FRPL counts: students certified by application or direct certification in non-CEP schools, and by direct certification or alternative income forms in CEP schools.[5] The second group includes 9 states that report all students in CEP schools, or 1.6

[3] C01 is aid distributed through the state's main formula; C06 is aid reported as targeted to disadvantaged or low-achieving students. For example, Texas's compensatory allotment is reported as C01, while Michigan's at-risk program is reported as C06.

[4] This matches over 90 percent of FRAC CEP schools each year. Most unmatched schools are private schools that are not in the CCD.

[5] The 22 states that report actual FRPL counts include 15 that use FRPL in funding: California, Colorado, Connecticut, Iowa,

times the identified student percentage, as FRPL-eligible.[6] The remaining states do not report FRPL or do not report it consistently over time.

For funding, I restrict the sample to 15 states that report actual FRPL counts to the CCD throughout the sample period and allocate poverty funding using FRPL, including California, Colorado, Connecticut, Iowa, Kansas, Kentucky, Maine, Michigan, Minnesota, New Jersey, New York, Rhode Island, Texas, Washington, and Wisconsin. These 15 states account for about half of enrollment in CEP-adopting districts nationally. In supplementary analyses, I also examine states that report actual FRPL counts but do not use them for funding, states that impute FRPL for CEP schools, and all states with general formula aid data.

I further restrict the sample to districts that adopt CEP by 2024–25. The treated group consists of the 1,248 districts that first adopt CEP between 2014–15 and 2019–20. The comparison group consists of districts that have not yet adopted in a given year: the treated cohorts before their own adoption and the 1,367 districts that adopt between 2020–21 and 2024–25, which are never treated during the sample period. I exclude districts that never adopt CEP, because many are ineligible and eligible non-adopters may differ from adopters in ways that affect funding trends. Nevertheless, adding eligible never-adopters to the comparison group yields similar estimates (Table 6, columns 3 and 4). I also exclude districts first observed in CEP in 2013–14, because the FRAC data begin that year and these districts may have adopted during the pilot phase, so their adoption year is unknown.

Table 1 reports summary statistics in 2012–13 for all districts and by year of first CEP adoption. Districts that adopt CEP during the sample period (2014–15 to 2019–20) are poorer than those that adopt later. They have higher FRPL shares, more non-White students and English learners, and larger enrollment. State general formula aid per student is similar across adoption cohorts, but earlier adopters receive less local and more federal revenue. By 2019–20, about 70 percent of enrollment in these districts is in CEP schools, on average. Because higher-poverty districts adopt CEP earlier, all specifications control for state indicators and 2012 SAIPE child poverty interacted with state, so that adopting districts are compared with not-yet-adopting districts in the same state with similar baseline poverty.

Kansas, Kentucky, Maine, Michigan, Minnesota, New Jersey, New York, Rhode Island, Texas, Washington, and Wisconsin. 7 that do not: Alabama, Arizona, Florida, Idaho, Illinois, Indiana, and Oregon.

[6]States that report all students in CEP schools as FRPL-eligible: Mississippi, Montana, Nevada, New Mexico, Pennsylvania, and South Carolina. States that report 1.6 times the identified student percentage: Georgia, South Dakota, and Virginia.

4 Empirical Strategy

4.1 Difference-in-Differences (DiD)

The goal is to estimate the effect of CEP adoption on access to free meals, meal applications, the reported FRPL share, and state funding using the DiD estimator developed by Callaway and Sant'Anna (2021). I compare treated districts with not-yet-treated districts in the same state with similar baseline poverty. A district is treated once at least one of its schools adopts CEP. Treated districts are those that first adopt CEP between 2014–15 and 2019–20. The comparison group consists of districts that have not yet adopted CEP in a given year. It includes two types of districts: (i) districts that adopt CEP later in the sample period, which serve as comparisons only until they adopt, and (ii) districts that first adopt between 2020–21 and 2024–25, which do not adopt during the sample period because the outcomes end in 2019–20. I control for state indicators and 2012 SAIPE child poverty rate interacted with the state indicators. This allows the relationship between baseline poverty and the outcome to differ by state.

The identifying variation comes from differences in the timing of adoption. The identifying assumption is conditional parallel trends. Within the same state and at similar levels of baseline poverty, the outcomes of districts that adopt earlier would have evolved like those of districts that adopt later had they not adopted CEP. The design absorbs shocks common to districts in a state in a given year (e.g., changes in base funding or the statewide adoption of Medicaid direct certification for school meals) and shocks that vary with baseline poverty differently in each state (e.g., a state's funding formula reform that directs more aid to poorer districts).

Let $G_i \in \{2014, \ldots, 2024\}$ denote the year district i first adopts CEP. For each treated cohort $g \in \{2014, \ldots, 2019\}$ and year $t \geq g$, the cohort-time average treatment effect on the treated is

$$ATT(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G_i = g] - \mathbb{E}[Y_t - Y_{g-1} \mid G_i > t], \tag{1}$$

where Y_{g-1} is the outcome in the year before cohort g adopts, the first term is the outcome change for cohort g, and the second is the change for districts not yet treated by year t. In practice, the second term is estimated conditional on state indicators and baseline poverty, so each cohort is compared with not-yet-treated districts in the same state with similar baseline poverty.

I aggregate the $ATT(g,t)$ into event-study estimates by event time $e = t - g$:

$$\theta(e) = \sum_{g} \mathbf{1}\{g + e \leq \tau\} P(G_i = g \mid G_i + e \leq \tau) ATT(g, g+e), \qquad e = -5, \ldots, 5, \tag{2}$$

where τ is the last sample year (2019–20), and pre-treatment estimates use $g-1$ as a common base period. The overall effect, θ^O, is the average of $\theta(e)$ over event times 0 through 5; the pre-treatment average is the average of $\theta(e)$ over event times -5 through -2. The unit of observation is a school district, and standard errors are clustered by district.

4.2 Funding Simulation

My second strategy passes the estimated change in the CCD FRPL share through each state's funding formula. I simulate the aid each district would receive if its state formula responded to the FRPL share reported to the CCD. For district d in state s in school year t, simulated aid per pupil is

$$\widetilde{A}_{dt} = w_{st}\, FRPL_{d,t-1}, \tag{3}$$

where $FRPL_{d,t-1}$ is the district's FRPL share in the prior school year (most formulas count students with a one-year lag) and w_{st} is the marginal state aid per FRPL-eligible pupil implied by the state's formula in that year. I take poverty weights and base amounts from EdBuild's FundEd state policy reports for 2016–2019 (EdBuild, 2016, 2018), supplemented by state documents.

I estimate equation (3) with the same event-study specification and districts as actual aid. Because treated and comparison districts in the same state face the same weight, the estimate equals the weight times CEP's effect on reported FRPL. This is the aid change implied by the measurement change alone. Comparing it with the estimate for actual aid shows how much of this implied change reaches districts. Standard errors are clustered by district, so they reflect only uncertainty in the FRPL response.

5 Results

5.1 Effects on Meal Access and Applications

I first examine how CEP adoption affects access to subsidized meals and household applications. Figure 3 plots the event-study estimates separately for 22 states that report actual FRPL counts to the CCD and 9 states that report imputed counts. Panel A shows the share of students eligible for subsidized meals. The pre-periods are flat and close to zero. After CEP adoption, the share jumps by about 21 percentage points in both sets of states and remains elevated through five years after adoption. Panel B depicts the share of students approved for subsidized meals by submitting an application. The pre-adoption coefficients for applications are small but positive and statistically significant. This indicates that the share approved for FRPL by application was declining faster in treated districts than in control districts prior to adoption, but the magnitudes are small relative to the post-adoption estimates. This pattern is consistent with selection into treatment. Districts with rising direct certification (hence less need for applications) are more likely to reach the CEP eligibility threshold. The share drops by about 15 percentage points at adoption but goes back up in later years.

Table 2 reports the corresponding DiD estimates, averaged over event times 0 to 5. Panel A shows the effects on meal access. Column 1 shows that in the 22 states that report actual FRPL counts, the share of students eligible for subsidized meals increases by 21.2 percentage points, or 37 percent of the pre-adoption mean of 57.1 percent. The estimates are similar in the subset of these states that use FRPL counts in funding (column 2), in states that report imputed counts (column 3), and in the SFAs linked to the districts of the funding analysis (column 4).

Table 2 Panel B, Column 1 shows that the share of students approved for subsidized meals by application declines by 13.7 percentage points, or 57 percent relative to the pre-adoption mean of 24.1 percent. The effects are similar in states that used FRPL in funding (column 2) and in states that report imputed counts (column 3), and are smaller in the linked sample (column 4).

Applications do not fall by 100 percent because of partial adoption and phase-in. First, some school districts adopt CEP for only a subset of their schools, so applications continue in the remaining schools. Moreover, some districts phase in CEP across schools over several years. [7] Taken together, CEP affects

[7] Among school food authorities linked to the school districts of the funding analysis, the decline is smaller, 9.8 percentage points, or 45 percent of the pre-adoption mean (column 4). The linked sample excludes school districts that could not be matched to a district in the funding analysis, and it uses the district's 2012 SAIPE child poverty rate rather than the 2013–14 directly certified

access and applications similarly in states that report actual and imputed counts.

5.2 Effects on Reported FRPL Shares Relative to Child Poverty

Next, I examine how CEP affects the FRPL data reported to the CCD. I estimate the effects separately for the 22 states that report actual FRPL counts and the 9 states that report all students in CEP schools, or 1.6 times the identified student percentage, as FRPL-eligible. In states that report actual counts, the reported FRPL share declines by 1.9 percentage points after adoption (Figure 4, Panel A, left; Table 3). The middle graph plots the log FRPL share in the CCD together with the log share of children aged 5–17 in poverty from SAIPE. After adoption, the reported FRPL share falls while child poverty rises slightly relative to not-yet-treated districts. The right graph shows the difference: reported FRPL falls by 4.9 log points relative to child poverty, averaged over years 0 to 5. Panel B shows the opposite pattern in states that impute FRPL counts. The reported FRPL share rises by 10.0 percentage points, much more than child poverty, so FRPL rises by 9.6 log points relative to poverty.

Table 3 reports the corresponding DiD estimates. Panel A, column 1 shows that in states that report actual counts, the reported FRPL share declines by 1.9 percentage points, or 3.0 percent relative to the pre-adoption mean. Columns 2–4 benchmark the FRPL share against SAIPE child poverty. The log FRPL share declines by 3.0 log points (column 2), while log child poverty increases by 1.9 log points (column 3). Relative to need, reported FRPL falls by 4.9 log points (column 4).[8]

Panel B of Table 3 restricts the sample to the 15 states that allocate poverty funding by FRPL. The decline is smaller in these states. The reported FRPL share falls by 1.2 percentage points, or 1.9 percent of the pre-adoption mean, and by 3.1 log points relative to need. In the seven states that report actual counts but do not use FRPL in funding, the reported FRPL share falls by 3.4 percentage points (column 3). This is consistent with stronger incentives to keep collecting household income information where FRPL is used in the funding formula.

In states that impute FRPL counts, the reported FRPL share instead rises by 10.0 percentage points, or 15 percent of the pre-adoption mean (Panel C, column 1), and by 9.6 log points relative to need (column 4). Child poverty rises by 3.1 log points.

share as the baseline covariate.

[8]Child poverty was already rising before adoption, so the pre-adoption estimates in columns 3 and 4 are statistically significant (-0.012 and 0.011 in Panel A; -0.016 and 0.014 in Panel B). The post-adoption estimates in column 4 have the opposite sign and are two to five times as large. In Panel C, the pre-adoption estimate for the reported FRPL share is also statistically significant (-0.011), but the estimate relative to need is close to zero (0.001).

Taken together, CEP lowers the CCD FRPL share relative to child poverty where states report actual counts and raises it where states impute counts. Meal applications fall by about 13 percentage points in both groups (Table 2), so these differences reflect how states report CEP schools rather than differences in take-up. As a result, the CCD FRPL share is no longer comparable across states or over time. Appendix Figure A.3 repeats the comparison using the ACS share of children below 185 percent of poverty. The results are similar to Figure 4.

5.3 Effects on State General Formula Aid

The key question is whether the decline in reported FRPL relative to poverty affects state K–12 funding. Figure 5 plots two sets of event-study estimates of state general formula aid per student. The solid line shows the DiD estimates of the effect of CEP on aid from equation (2), and the dashed line shows formula-simulated aid from equation (3), the aid each district would receive if its state formula responded only to the reported FRPL share. Before adoption, both sets show flat trends. After adoption, the DiD estimates hover around zero, and their confidence intervals widen in later years. Simulated aid declines from year 2, by about $40 per student. These declines lie within the DiD confidence intervals.

Table 4, column 1, shows that CEP adoption has no detectable effect on general formula aid. The estimate is $7 per student, or 0.1 percent of the pre-adoption mean, and is not statistically significant. The 95 percent confidence interval rules out losses larger than $105 and gains larger than $120. The formula-simulated loss is $42 per student, or 0.6 percent of the mean (column 2). Although the simulated loss is small relative to total general formula aid, it is larger relative to the poverty funding that formulas are designed to target: about 3 percent of the $1,265 per student that treated districts would receive through FRPL-based weights before adoption. Taken together, the results rule out large losses in state aid from the change in FRPL data.

5.4 Heterogeneity

By State

To explore how the effects vary across states, Figure 6 reports state-specific estimates. The left panel shows effects on the reported FRPL share in the CCD. The decline is concentrated in three states: Connecticut

(−9.9 percentage points), Minnesota (−5.8), and New York (−3.2).[9] Texas also shows a small but statistically significant decline (−0.9), while the estimates for the remaining states, including Michigan, are close to zero and not statistically significant.

The right panel shows DiD and simulated effects on state general formula aid per student. Formula-simulated losses are also concentrated in Connecticut, Minnesota, and New York, the states that combine large declines in reported FRPL with relatively large poverty weights. The state-specific DiD estimates are noisy, and most cannot distinguish the simulated loss from zero. Connecticut is the exception: actual aid falls by $833 per student, close to the simulated loss of $648.[10] In Minnesota and New York, the point estimates are negative (−$120 and −$77) and consistent with both the simulated losses and zero. Taken together, the declines in reported FRPL vary substantially across states and line up with the formula-implied losses, but the state-level funding estimates are too imprecise to confirm them state by state.

By Treatment Intensity

I split treated districts into district-wide adopters and partial adopters in Table 5. The decline in the reported FRPL share is similar in both groups. The funding estimates do not follow this pattern. General formula aid increases by a statistically insignificant $85 per student for district-wide adopters and falls by $148 for partial adopters, and the difference is statistically significant ($p = 0.02$). The simulated losses are similar for the two groups ($33 and $48, $p = 0.50$). Because the change in reported FRPL does not differ between the groups, the FRPL channel cannot explain this difference. The loss for partial adopters is about three times its simulated loss, and the two groups differ before adoption: partial adopters have lower FRPL shares and receive less formula aid. The difference in actual aid therefore more likely reflects other differences between the groups than the change in FRPL data.

5.5 Robustness Checks

Different Control Groups, Samples, Estimators and Inference

Table 6 presents robustness checks for the reported FRPL share (Panel A) and general formula aid per student (Panel B) using different controls, comparison groups, samples, estimators, and inference. Column

[9] New Jersey (−5.9 percentage points, standard error 3.4) is omitted because it has only four comparison districts.

[10] Connecticut has only 26 treated and 15 comparison districts, and its pre-adoption estimate is positive ($177, standard error $74). Because the state distributed block grants and did not use its formula until 2018–19, the simulated loss comes entirely from the last two years of the sample, when the DiD estimate may also reflect the return to the formula.

1 shows the main estimates, which control for state indicators and district baseline child poverty interacted with the state indicators and use data through school years 2019–20. Column 2 controls for baseline child poverty with a single slope across states instead of interacting it with the state indicators. Columns 3 and 4 add districts that never adopt CEP but were eligible to the comparison group. Column 3 adds districts with at least one eligible school, and column 4 adds districts that are eligible for CEP district-wide. Column 5 restricts the treated districts to the 2014–2016 cohorts and uses a balanced event window from -3 to 3. Columns 6 and 7 change the weighting and the estimator. Column 6 weights districts by enrollment, and column 7 uses the doubly robust estimator. Column 8 reports wild cluster bootstrap standard errors with clusters by state.

The decline in the reported FRPL share is robust across specifications, ranging from 0.9 to 1.5 percentage points. All estimates are statistically significant at the 1 percent level except the enrollment-weighted estimate (column 6), which is similar in magnitude but less precise. The funding estimates are not stable, and none is statistically significant at the 5 percent level. The point estimates range from a loss of $137 per student (column 6) to a gain of $19 (column 7).

Robustness to State Formula Reform

The main specification absorbs statewide shocks, such as a formula reform that changes aid for all districts in a state, and, by interacting baseline poverty with the state indicators, reforms whose effects differ with district poverty within a state. A specification with a single poverty slope for all states cannot fully absorb a reform that directs new aid to high-poverty districts in one state. Because early CEP adopters are high-poverty districts, such a reform could be confounded with adoption. To check this, columns 2–4 of Table 8 drop states that changed their funding formulas during the sample period. Dropping these states leaves the decline in the reported FRPL share similar (1.2 to 1.4 percentage points), and the estimated effect on general formula aid remains close to zero ($-$\$10 to $+$\$16 per student). In column 5, I enter SAIPE child poverty with a single slope for all states. The decline in the reported FRPL share remains (1.5 percentage points), but general formula aid now falls by $98.8 per student, significant at the 10 percent level. The loss under a common poverty slope therefore reflects state-specific changes that favored poorer districts, which the main specification absorbs.

Placebo Tests

In addition, I conduct a series of placebo tests in Table 9. Columns 1–4 examine state revenues that do not depend on FRPL: special education, transportation, career and technical education, and other state aid. None of these estimates is statistically significant, and all are small relative to their means. Columns 5–8 examine child nutrition revenues, which do depend on meal counts and therefore serve as a check rather than a placebo. As expected, CEP raises federal child nutrition revenue by $85 per student (17 percent) and reduces local revenue from meal sales by $21 (15 percent), increasing total child nutrition revenue by $66 per student (10 percent). This is consistent with Rothbart, Schwartz and Gutierrez (2023).[11] Column 9 examines federal Title I revenue, which is allocated using Census poverty counts rather than FRPL and should therefore not respond to CEP. Title I revenue declines by $15.2 per student (3.1 percent) after adoption, but the pre-adoption coefficient is positive and statistically significant ($12.2), indicating that Title I revenue was already declining in adopting districts relative to comparison districts. This outcome therefore does not provide a clean placebo.

Check CCD Against State Sources

In Appendix Figure A.4, I compare the FRPL counts reported to the CCD with the counts published by state departments of education in California, Pennsylvania, Texas, and Virginia. The two series match closely in California, Texas, and Virginia. Pennsylvania reports all students in CEP schools as FRPL-eligible to the CCD, so its CCD share relative to child poverty rises by about 18 log points after adoption. In contrast, the state's own count of low-income students relative to child poverty declines by 7 log points, similar to the decline in states that report actual counts.

Robustness to Violation of Parallel Trends

The main estimates assume that, without CEP, adopting districts would have followed the same trends as not-yet-adopting districts. In Appendix Figure A.5, I use the approach of Rambachan and Roth (2023) to check robustness to violation of parallel trends. The approach allows the trends of the two groups to diverge after adoption by up to $\bar{M}$ times the largest year-to-year divergence observed before adoption. The decline in the FRPL share relative to child poverty remains statistically significant if the post-adoption divergence

[11] The pre-adoption coefficients for federal and local child nutrition revenue are statistically significant ($-$15 and $6) but small relative to the post-adoption estimates.

is at most 10 percent of the largest pre-adoption divergence ($\bar{M} = 0.1$), but not at 20 percent. The decline in reported FRPL is therefore robust to small, but not large, violations of parallel trends. The estimate for general formula aid is not statistically significant under any value of $\bar{M}$.

6 Conclusion

This paper examines how CEP affects the free and reduced-price meal data that many states use to direct funding toward low-income students. Using linked administrative data, I find that CEP adoption reduces applications across the board. As a result, FRPL data become a less reliable proxy for student poverty. Reported FRPL shares diverge from child poverty in opposite directions depending on how states count students in CEP schools. Relative to poverty, FRPL shares fall by 5 log points in states that report actual counts and rise by 10 log points in states that report all students in CEP schools, or a multiple of the identified student percentage, as eligible. Despite these changes, I find little evidence of funding loss. In the 15 states that allocate poverty funding by FRPL, state formulas imply a loss of $42 per student, and I can rule out annual losses in state formula aid exceeding $105 per student.

The losses are small because poverty weights account for a small share of total formula aid. They are concentrated in Connecticut, Minnesota, and New York, the states where large declines in reported FRPL meet relatively large poverty weights. Relative to the poverty funding that formulas are designed to target, however, the implied loss is about 3 percent, and the state-level estimates are too imprecise to rule out larger losses in these states.

More broadly, the results show that once CEP spreads, reported FRPL reflects state reporting rules as much as student need. CEP can lower reported FRPL shares in one state and raise them in another. This affects any use of FRPL as a poverty measure, in state funding formulas and in research. The problem is likely to grow as CEP continues to expand. States that continue to rely on FRPL will need to monitor it closely or move toward alternative measures to avoid unintentionally cutting funding for low-income students.

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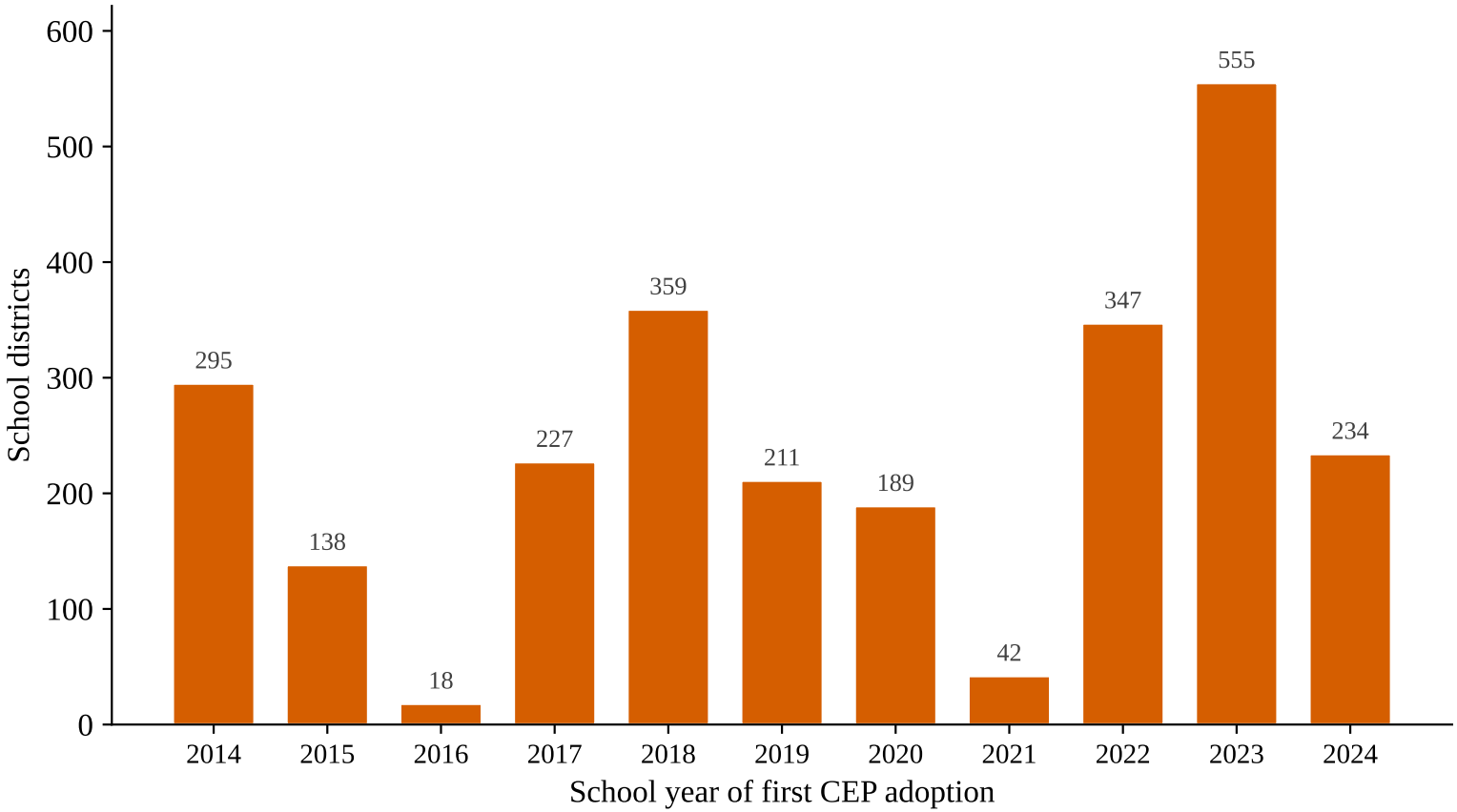
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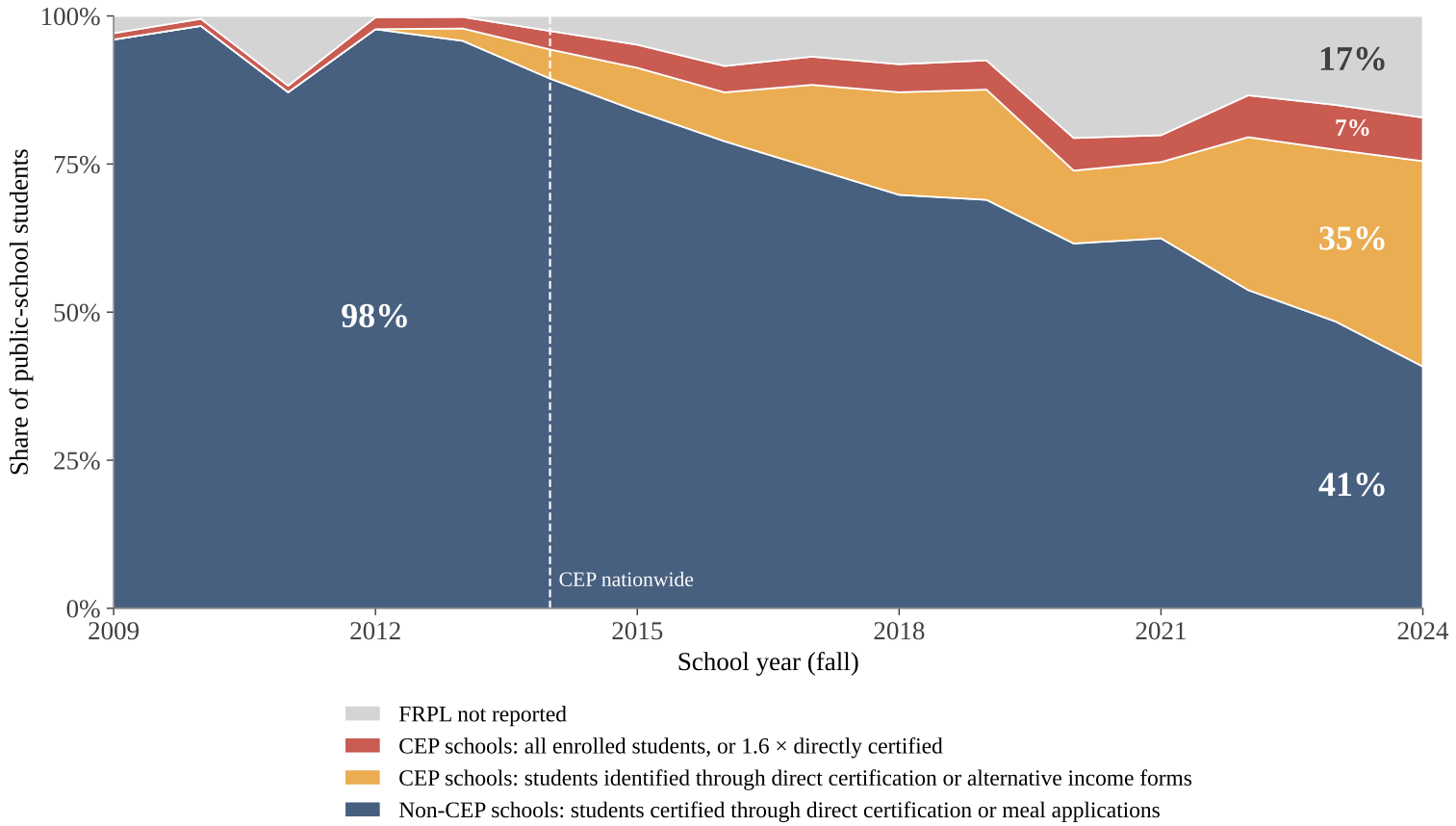
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Figure 1: Year of First CEP Adoption



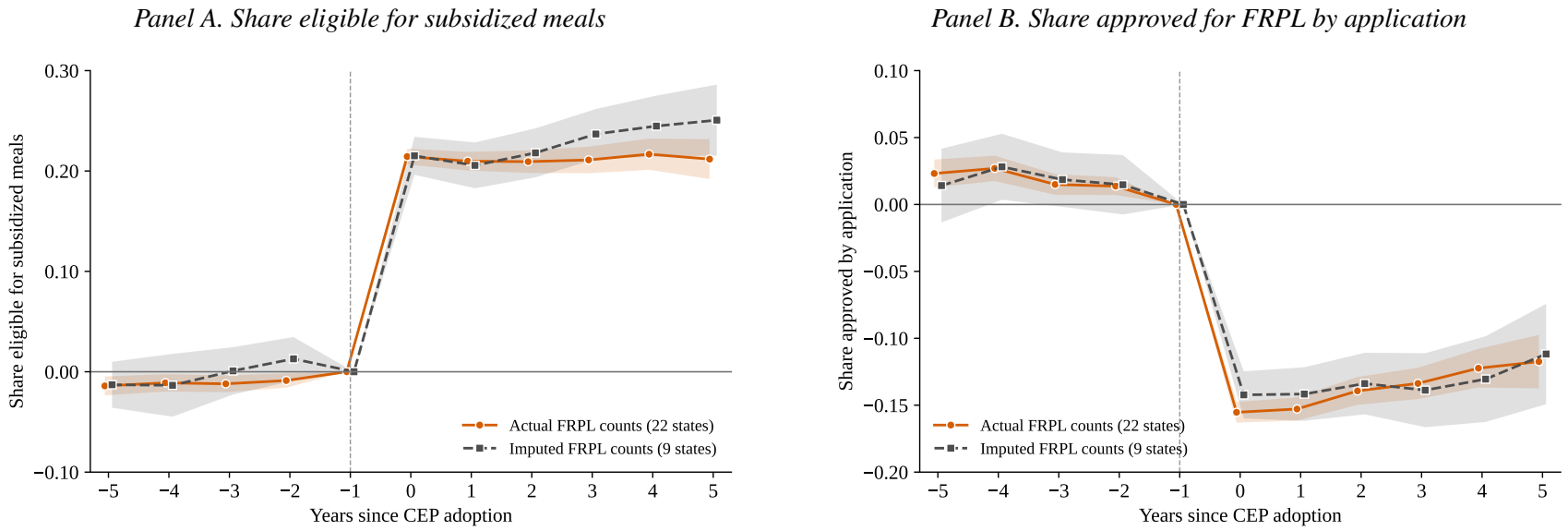
Notes: The figure shows the number of school districts by the school year in which at least one school in the district first adopts CEP. The sample includes districts in the 15 states that allocate poverty funding by FRPL and report actual FRPL counts to the CCD. Because the sample period is 2009–10 to 2019–20, school districts that adopt in 2020–2024 are never treated in the sample period.

Figure 2: How School Districts Report Students Eligible for Free or Reduced-Price Meals to CCD



Notes: Share of public-school enrollment by how each school's count of students eligible for free or reduced-price meals (FRPL) is reported to the Common Core of Data (CCD), weighted by enrollment. Non-CEP schools: students certified through direct certification or meal applications. CEP schools, students identified: students identified through direct certification or alternative income forms, as reported by the state. CEP schools, all enrolled: FRPL reported as at least 99 percent of enrollment, or as 1.6 times the identified student percentage (Georgia and Virginia); includes a few non-CEP schools, mostly Provision 2 and 3, that report 100 percent. FRPL not reported: mostly states that stopped reporting FRPL (Delaware, the District of Columbia, Illinois, Massachusetts, Oklahoma, Tennessee, West Virginia) and California in 2011–12. The dashed line marks 2014–15, the first year CEP was available nationwide.

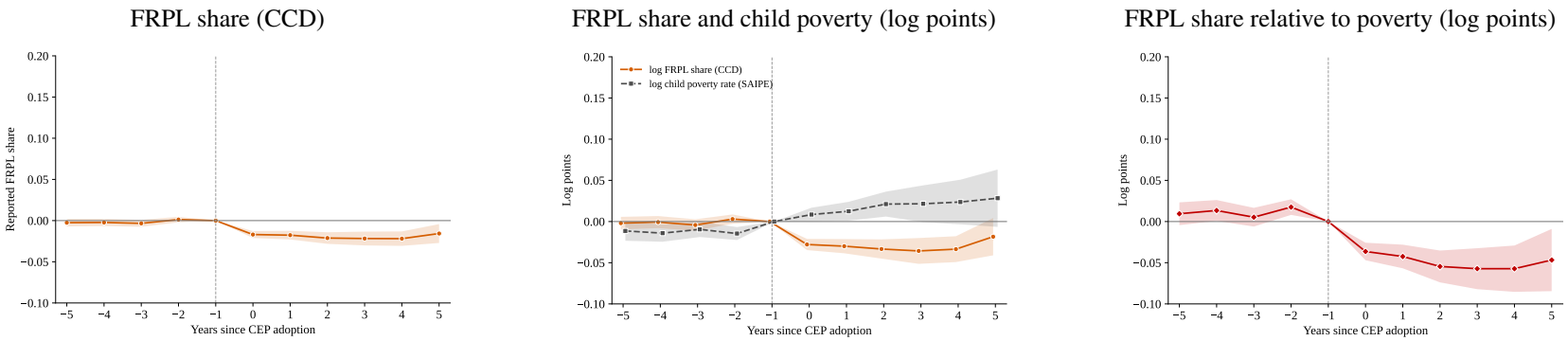
Figure 3: Effects of CEP on Meal Access and Applications by CCD Reporting Rule



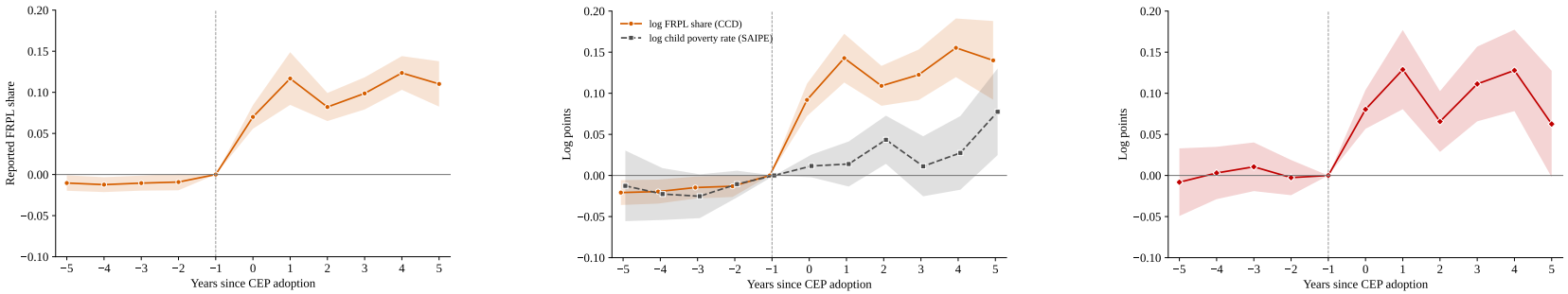
Notes: This figure shows the event-study estimates using equation (2). The shaded areas are 95 percent confidence intervals, standard errors clustered by school district. The sample includes all public school food authorities that adopt CEP by 2024-25, in states that report actual FRPL counts to the CCD (22 states) and in states that report all students in CEP schools, or 1.6 times the identified student percentage, as FRPL eligible (9 states).

Figure 4: Effects of CEP on Reported FRPL Shares Relative to Child Poverty, by CCD Reporting Rule

Panel A. States that report actual FRPL counts (22 states)

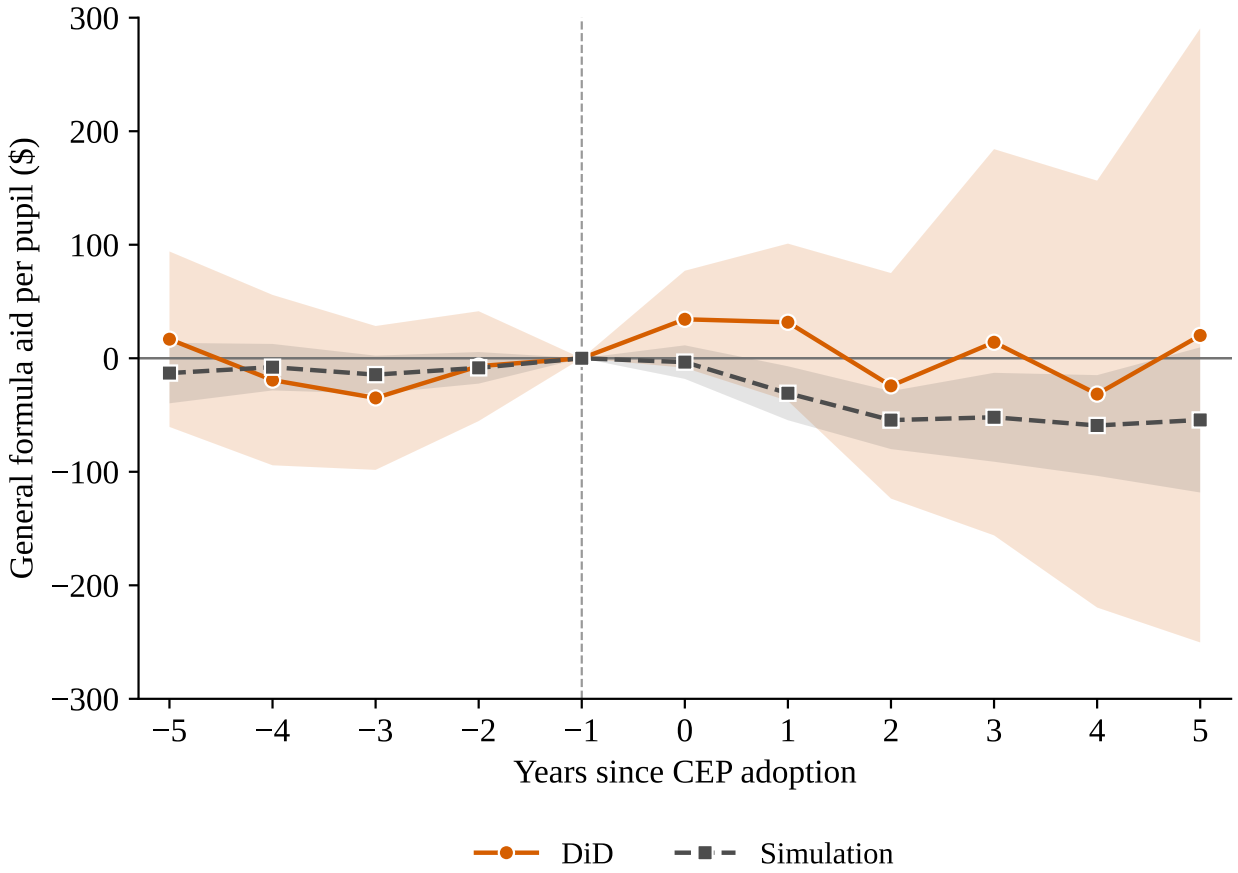


Panel B. States that report imputed FRPL counts (9 states)



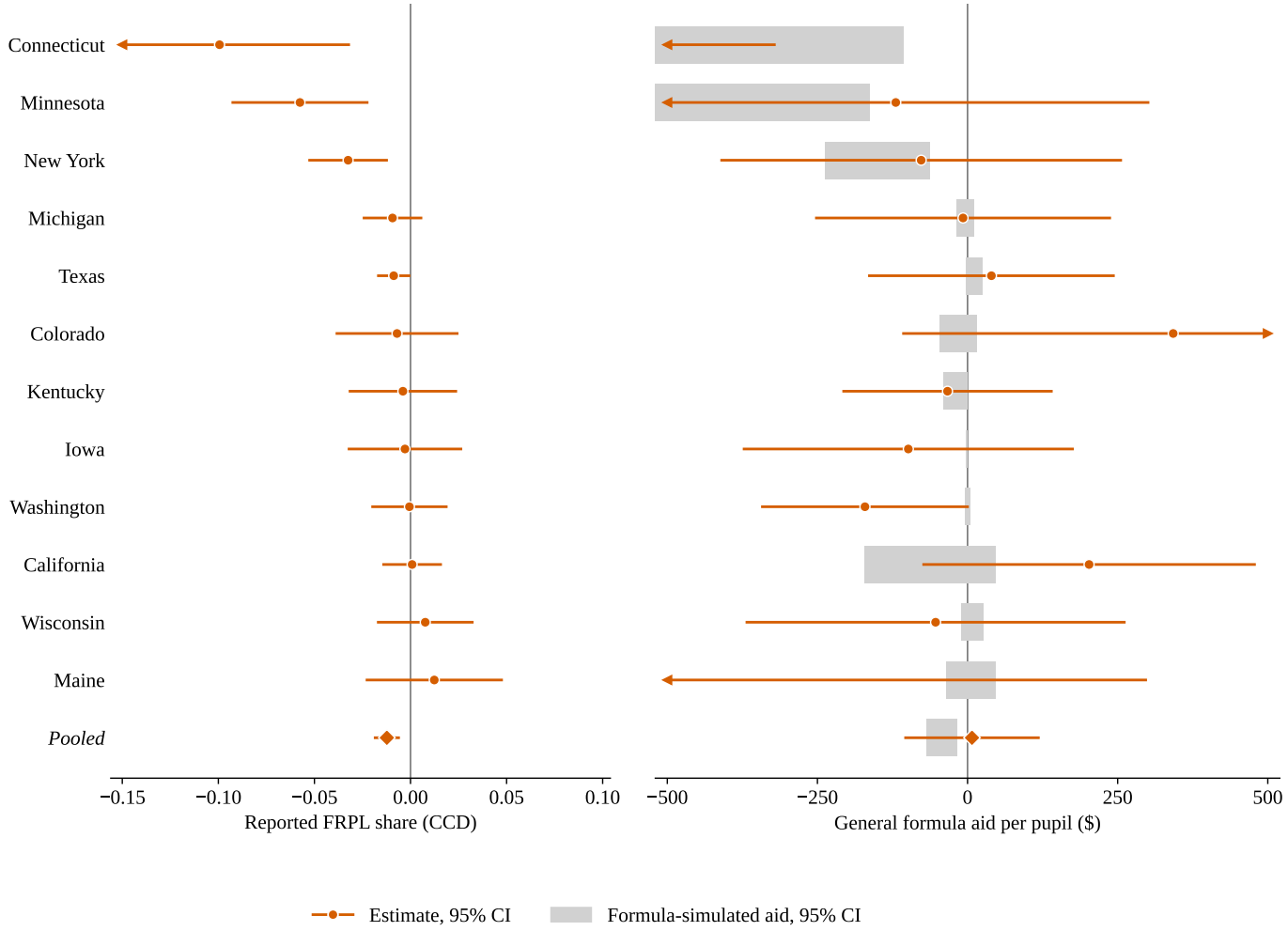
Notes: This figure shows event-study estimates from equation (2). Shaded areas are 95 percent confidence intervals, with standard errors clustered by school district. Panel A includes the 22 states that report actual FRPL counts to the Common Core of Data (CCD). Panel B includes the 9 states that report all students in CEP schools, or 1.6 times the identified student percentage, as FRPL eligible. The outcome in the left graph is the FRPL share reported to the CCD. The outcomes in the middle graph are the log FRPL share and the log share of children aged 5–17 in poverty from the Census Bureau's Small Area Income and Poverty Estimates (SAIPE), on the same district-years. The outcome in the right graph is the difference between the two series.

Figure 5: Effects of CEP on State Formula Aid



Notes: This figure plots two sets of event-study estimates of state general formula aid per student. The solid line shows the DiD estimates of the effect of CEP on aid from equation (2), and the dashed line shows formula-simulated aid from equation (3). The shaded areas are 95 percent confidence intervals.

Figure 6: Effects of CEP by State



Notes: The left panel shows reported FRPL share (Common Core of Data). The right panel shows the effects on general formula aid (F-33 item C01) per student. The solid line shows the DiD estimates of the effect of CEP on aid from equation (2), and the grey shade shows formula-simulated aid from equation (3). Shaded areas are 95 percent confidence intervals, with standard errors clustered by school district.

Table 1: Summary Statistics

	All	Year of first CEP adoption		
		2014–2016	2017–2019	2020–2024
	(1)	(2)	(3)	(4)
Students				
FRPL share	0.55	0.67	0.62	0.47
Child poverty rate	0.22	0.28	0.25	0.18
Share non-White	0.38	0.50	0.46	0.30
Share Black	0.06	0.10	0.07	0.04
Share Hispanic	0.26	0.34	0.33	0.19
Share English learners	0.08	0.11	0.11	0.06
Enrollment	5,373	9,909	5,334	3,855
Revenue per pupil ($)				
Total	18,428	17,838	17,558	19,134
State	9,512	9,830	9,676	9,309
General formula aid	6,744	6,962	6,952	6,550
Local	7,479	5,967	6,300	8,676
Federal	1,438	2,041	1,582	1,149
Child nutrition	409	534	467	332
CEP				
Share of enrollment in CEP schools, 2019–20	0.35	0.74	0.70	0.02
School districts	2,615	451	797	1,367
States	15	15	15	15

Notes: This table shows the unweighted means across school districts in the 2012–13 school year. The sample includes districts in the 15 states that allocate poverty funding by FRPL and report actual FRPL counts to the CCD. FRPL share is FRPL students reported to the CCD over enrollment. Child poverty rate is the share of children aged 5–17 below 100 percent of the federal poverty line in the Census Bureau's Small Area Income and Poverty Estimates (SAIPE). Dollars in 2024 dollars.

Table 2: Effects of CEP on Free-Meal Access and Applications by CCD Reporting Rule

	Actual counts (1)	Actual, used in funding (2)	Imputed counts (3)	funding districts (4)
Panel A. Eligible for subsidized meals (%)				
Pre-CEP	-1.15^{***}	-1.30^{***}	-0.32	-0.46
Years -5 to -2	(0.31)	(0.34)	(0.88)	(0.40)
Post-CEP	21.22^{***}	20.89^{***}	22.85^{***}	21.32^{***}
Years 0–5	(0.52)	(0.61)	(1.06)	(0.71)
Mean before CEP	57.1	56.0	55.3	52.4
Post-CEP, % of mean	37.2	37.3	41.3	40.7
Panel B. Approved for FRPL by application (%)				
Pre-CEP	1.97^{***}	1.87^{***}	1.89^{**}	0.83^{**}
Years -5 to -2	(0.33)	(0.36)	(0.77)	(0.39)
Post-CEP	-13.68^{***}	-13.47^{***}	-13.31^{***}	-9.84^{***}
Years 0–5	(0.51)	(0.64)	(1.13)	(0.60)
Mean before CEP	24.1	24.5	20.3	21.9
Post-CEP, % of mean	-56.8	-55.1	-65.5	-44.9
School food authorities	5,007	3,689	1,144	2,531
States	22	15	9	15

Notes: This table shows the event-study estimates from equation (2). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Effects of CEP on Reported FRPL Shares and Need, by CCD Reporting Rule

	(1) Reported FRPL share (CCD)	(2) Log FRPL share (CCD)	(3) Log child poverty rate (SAIPE)	(4) FRPL share relative to need (2) − (3)
Panel A. Actual FRPL counts (22 states)				
Pre-CEP	−0.002	−0.001	−0.012***	0.011**
Years −5 to −2	(0.002)	(0.003)	(0.004)	(0.005)
Post-CEP	−0.019***	−0.030***	0.019**	−0.049***
Years 0–5	(0.003)	(0.005)	(0.008)	(0.009)
Mean before CEP	0.635			
School districts	3,472	3,472	3,472	3,472
Panel B. Actual FRPL counts, FRPL used in funding (15 states)				
Pre-CEP	−0.002	−0.002	−0.016***	0.014**
Years −5 to −2	(0.002)	(0.003)	(0.005)	(0.006)
Post-CEP	−0.012***	−0.016**	0.014	−0.031***
Years 0–5	(0.003)	(0.007)	(0.011)	(0.011)
Mean before CEP	0.625			
School districts	2,615	2,615	2,615	2,615
Panel C. Imputed FRPL counts (9 states)				
Pre-CEP	−0.011***	−0.017***	−0.018	0.001
Years −5 to −2	(0.004)	(0.006)	(0.014)	(0.014)
Post-CEP	0.100***	0.127***	0.031**	0.096***
Years 0–5	(0.008)	(0.012)	(0.013)	(0.016)
Mean before CEP	0.660			
School districts	773	773	773	773
Units	Share	Log points	Log points	Log points

Notes: This table shows the event-study estimates from equation (2). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Effects of CEP on State Formula Aid

	(1) General formula aid per pupil ($)	(2) Formula-simulated aid per pupil ($)
Pre-CEP	−11.1	−10.9
Years −5 to −2	(29.7)	(7.7)
Post-CEP	7.4	−42.4***
Years 0–5	(57.5)	(13.0)
Mean before CEP	7,559	7,559
Post-CEP, % of mean	0.10	−0.56
School districts	2,615	2,615
States	15	15

Notes: This table shows the event-study estimates of state general formula aid per student from equation (2) and of formula-simulated aid from equation (3). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Effects of CEP by Treatment Intensity

	(1) District-wide adoption	(2) Partial adoption
Panel A. Reported FRPL share (CCD)		
Pre-CEP	−0.002	−0.003
Years −5 to −2	(0.003)	(0.002)
Post-CEP	−0.012**	−0.010***
Years 0–5	(0.005)	(0.003)
Mean before CEP	0.678	0.547
Post-CEP, % of mean	−1.77	−1.83
p-value, post-CEP (1) = (2)	0.73	
Panel B. General formula aid per pupil, DiD ($)		
Pre-CEP	−0.5	−31.1
Years −5 to −2	(43.9)	(29.9)
Post-CEP	84.6	−148.4**
Years 0–5	(76.1)	(60.5)
Mean before CEP	8,117	6,716
Post-CEP, % of mean	1.04	−2.21
p-value, post-CEP (1) = (2)	0.02	
Panel C. Formula-simulated aid per pupil ($)		
Pre-CEP	−2.7	−21.2***
Years −5 to −2	(9.6)	(7.1)
Post-CEP	−33.3*	−47.7***
Years 0–5	(17.5)	(12.4)
Mean formula aid before CEP	8,117	6,716
Post-CEP, % of mean	−0.41	−0.71
p-value, post-CEP (1) = (2)	0.50	
Treated districts	768	478
School districts	2,135	1,845
States	15	15

Notes: This table shows the event-study estimates from equation (2) and of formula-simulated aid from equation (3). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Robustness of the Main Estimates

		Covariates	Adds eligible never-adopters		Sample	Estimator		Inference
	(1) Main	(2) Common poverty slope	(3) Any school eligible	(4) District eligible	(5) Balanced −3 to 3	(6) Enrollment-weighted	(7) Doubly robust	(8) Wild bootstrap
Panel A. Reported FRPL share (CCD)								
Pre-CEP	−0.002	−0.003*	−0.002	−0.002	−0.001	−0.011	−0.003	−0.002
Years −5 to −2	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)	(0.007)	(0.002)	(0.004)
Post-CEP	−0.012***	−0.015***	−0.013***	−0.013***	−0.009**	−0.012	−0.012***	−0.012***
Years 0–5	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)	(0.007)	(0.003)	(0.005)
School districts	2,615	2,615	3,415	3,064	1,818	2,615	2,615	2,615
Panel B. General formula aid per pupil ($)								
Pre-CEP	−11.1	−4.4	−26.1	−20.5	62.5	−17.8	−39.0	−11.1
Years −5 to −2	(29.7)	(29.2)	(27.7)	(27.8)	(65.1)	(55.4)	(34.4)	(22.7)
Post-CEP	7.4	−98.8*	−38.0	−20.7	−61.7	−136.9	18.5	7.4
Years 0–5	(57.5)	(54.6)	(49.2)	(50.5)	(48.7)	(93.4)	(55.8)	(82.3)
School districts	2,615	2,615	3,415	3,064	1,818	2,615	2,615	2,615
State indicators	Y	Y	Y	Y	Y	Y	Y	Y
SAIPE child poverty × state	Y		Y	Y	Y	Y	Y	Y
SAIPE child poverty		Y						
Eligible never-adopters			Y	Y				
Balanced event window					Y			
Enrollment-weighted						Y		
Propensity-score weighting							Y	
Wild bootstrap by state								Y

Notes: This table shows the event-study estimates from equation (2) in the 15 states that allocate poverty funding by FRPL and report actual FRPL counts to the CCD. Column (1): main specification, with state indicators and 2012 SAIPE child poverty interacted with the state indicators as covariates, unweighted, standard errors clustered by district, school years 2009–10 to 2019–20. Pre-CEP and post-CEP rows report the mean of event-time effects relative to year −1. Column (2) enters SAIPE child poverty with a single slope for all states. Columns (3) and (4) add districts that never adopt CEP by 2024–25 but were eligible to the comparison group: in column (3), districts with at least one school with an identified student percentage (ISP) of at least 0.40 in any year from 2014–15 to 2019–20; in column (4), districts whose listed schools had an enrollment-weighted ISP of at least 0.40, which could adopt CEP district-wide (FRAC school lists). Column (5) restricts the treated districts to the 2014–2016 cohorts, which are observed for three years after adoption, and reports event times −3 to 3 (pre-CEP: years −3 and −2; post-CEP: years 0–3). Column (6) weights districts by enrollment; column (7) uses the doubly robust estimator (inverse propensity weighting with an outcome regression). Column (8) reports wild cluster bootstrap standard errors with clusters by state. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Effects of CEP Outside the Main Sample

	Imputed FRPL counts		Actual FRPL counts		
	(1) FRPL used in funding	(2) FRPL not used in funding	(3) FRPL not used in funding	(4) Excl. Illinois	(5) National
Panel A. Reported FRPL share (CCD)					
Pre-CEP	-0.013^{***}	-0.010^{*}	0.000	0.003	
Years -5 to -2	(0.004)	(0.006)	(0.004)	(0.005)	
Post-CEP	0.069^{***}	0.112^{***}	-0.034^{***}	-0.032^{***}	
Years 0–5	(0.010)	(0.010)	(0.005)	(0.006)	
Mean before CEP	0.670	0.656	0.664	0.665	
Post-CEP, % of mean	10.30	17.08	-5.08	-4.80	
School districts	202	571	857	518	
Panel B. General formula aid per pupil ($)					
Pre-CEP	-72.2^{*}	17.6	-60.2^{*}	-36.9	-21.7
Years -5 to -2	(41.4)	(62.0)	(31.6)	(38.5)	(18.2)
Post-CEP	9.9	124.2	107.0^{**}	22.8	72.1
Years 0–5	(56.1)	(98.5)	(53.4)	(58.4)	(77.7)
Mean before CEP	5,474	6,105	5,692	6,357	6,863
Post-CEP, % of mean	0.18	2.03	1.88	0.36	1.05
School districts	202	571	857	518	5,274
States	3	6	7	6	48

Notes: This table shows the event-study estimates from equation (2) $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Robustness to State Funding Formula Reforms

		Excluding states with formula reforms			
	(1) Main	(2) Drop CA	(3) Drop CA, KS, CT	(4) Drop CA, KS, CT, TX	(5) Common poverty slope
Panel A. Reported FRPL share (CCD)					
Pre-CEP	−0.002	−0.003	−0.004*	0.002	−0.003*
Years −5 to −2	(0.002)	(0.002)	(0.002)	(0.003)	(0.002)
Post-CEP	−0.012***	−0.014***	−0.012***	−0.014***	−0.015***
Years 0–5	(0.003)	(0.004)	(0.004)	(0.005)	(0.004)
Mean before CEP	0.625	0.601	0.602	0.565	0.625
School districts	2,615	2,131	2,050	1,526	2,615
Panel B. General formula aid per pupil ($)					
Pre-CEP	−11.1	12.4	8.4	−0.7	−4.4
Years −5 to −2	(29.7)	(34.9)	(36.1)	(46.4)	(29.2)
Post-CEP	7.4	−9.9	16.3	9.1	−98.8*
Years 0–5	(57.5)	(61.6)	(63.3)	(79.3)	(54.6)
Mean before CEP	7,559	7,729	7,777	8,505	7,559
Post-CEP, % of mean	0.10	−0.13	0.21	0.11	−1.31
School districts	2,615	2,131	2,050	1,526	2,615
States	15	14	12	11	15

Notes: This table shows the event-study estimates from equation (2). Column (1): main estimates, districts in the 15 states that allocate poverty funding by FRPL and report actual FRPL counts to the CCD (Tables 3 and 4). Columns (2)–(4) drop states that changed their funding formulas during the sample period: California (Local Control Funding Formula from 2013–14, phased in through 2017–18); Kansas (block grants in 2015–16 and 2016–17 and a new formula from 2017–18); Connecticut (block grants until 2018–19); and Texas (new compensatory education weights in 2019–20). Column (5): all 15 states, with state indicators and 2012 SAIPE child poverty as covariates, so that the relationship between baseline poverty and the change in the outcome is the same in every state. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Effects of CEP on Other School Revenue

	State aid not tied to FRPL (placebo)				Child nutrition revenue				Federal aid (SAIPE-based)
	(1) Special education	(2) Transportation	(3) Career and technical	(4) Other state aid	(5) Federal	(6) State	(7) Local (meal sales)	(8) Total	(9) Title I
Pre-CEP	2.6	−11.7	0.0	25.7	−15.0***	−1.0	5.6***	−9.9***	12.2**
Years −5 to −2	(9.3)	(13.9)	(1.4)	(24.5)	(2.2)	(0.6)	(1.4)	(2.8)	(4.8)
Post-CEP	18.9	7.8	1.4	4.8	84.6***	1.3*	−20.9***	65.6***	−15.2*
Years 0–5	(14.6)	(12.7)	(1.4)	(44.0)	(5.4)	(0.7)	(3.5)	(6.8)	(9.0)
Mean before CEP	1,284	316	19	2,556	485	24	143	653	489
Post-CEP, % of mean	1.5	2.5	7.5	0.2	17.4	5.4	−14.7	10.0	−3.1
School districts	2,615	2,615	2,615	2,615	2,615	2,615	2,615	2,615	2,615

Notes: This table shows the event-study estimates from equation (2). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A Appendix

A.1 Simulation

The simulation asks how much state aid district would lose after CEP adoption if its state formula allocated poverty funding using the FRPL share the state reports to the CCD. Comparing the simulated path with the estimated aid response shows whether the observed change in aid is of the size and timing that the measurement change would produce. The simulation uses the districts in states that report actual FRPL counts to the CCD: 1,248 treated and 1,367 comparison districts in 15 states (CA, CO, CT, IA, KS, KY, ME, MI, MN, NJ, NY, RI, TX, WA, WI). These are the same districts and school years (2009–10 to 2019–20) used for the actual funding estimates.

For district d in state s in school year t,

$$\widetilde{A}_{dt} = g_{st}(FRPL_{d,t-1}), \tag{4}$$

where $FRPL_{d,t-1}$ is the district's cleaned CCD FRPL share in the prior school year. In most states g_{st} is linear, $g_{st}(x) = w_{st}\,x$, with w_{st} the marginal state aid per FRPL-eligible pupil. Two states are nonlinear:

$$\begin{aligned} &\text{California:} \quad g(x) = B\left[0.2\,x + 0.5\max(0, x - 0.55)\right], \quad t \geq 2013\text{–}14, \\ &\text{Minnesota:} \quad g(x) = U\,x\min(1, x/0.8), \end{aligned}$$

where B is the LCFF base grant and $U = 0.6\,(\text{base} - \$839)$ is Minnesota's compensatory revenue per pupil unit.

Table A.1 lists the formula parameters. Poverty weights come from EdBuild's FundEd "Poverty Funding Policies" reports and base amounts from its "Base Amount Policies" reports, both read from archived versions dated 2016 to 2019 (EdBuild, 2018, 2016). Where EdBuild does not report a parameter I use state sources: Texas basic allotments by year from the Texas Education Agency (Texas Education Agency, 2024), Michigan's at-risk proration from the House Fiscal Agency (Michigan House Fiscal Agency, 2018), and the dates of Kansas's block grants from the Kansas Legislative Research Department (Kansas Legislative Research Department, 2021). Nominal amounts are converted to 2024 dollars with the ratio of the 2024 CPI-U annual average to the CPI-U annual average of the calendar year in which the reported fiscal year ends.

I estimate simulated aid as an outcome with the same specification from equation (2).

Table A.1: Formula parameters used in the simulation

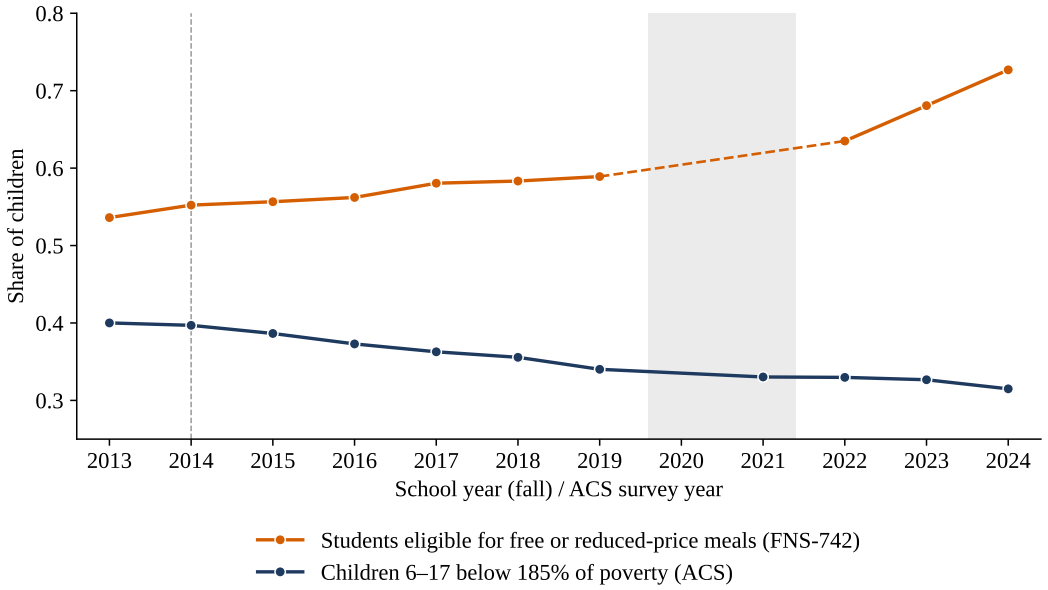
State	Rule (source year)	w_{st} (2024 $)
CA	0.2× base, plus 0.5× base per pupil above 55%; base $7,193–$8,712, midpoint $7,950 (FY2018); 0 before 2013–14	nonlinear
CO	0.12× base $6,367.90 (FY2017); free lunch only	978
CT	Block grants, formula not in use through FY2018; from FY2019 0.30× foundation $11,525	0 / 4,242
IA	0.0048× base $6,591 (FY2017)	40
KS	Block grants 2015–16 and 2016–17; otherwise at-risk 0.484× BASE $4,569 (FY2021); free lunch only	0 / 2,561
KY	0.15× base $3,981 (FY2016–FY2018); free lunch only	780
ME	0.15× base, midpoint of $5,134–$7,353 (FY2018)	1,170
MI	0.115× foundation $8,289 (FY2018), prorated to 78.1%	930
MN	$U = 0.6 \times (\$6{,}188 - \$839)$ (FY2018) per pupil unit, times concentration factor	nonlinear
NJ	0.46× base $11,009 (FY2017; range 0.41–0.46 by concentration)	6,482
NY	Pupil Need Index adds 0.65× FRL share; foundation $6,422 (FY2018)	5,214
RI	0.40× base $9,163 (FY2018)	4,578
TX	0.2× basic allotment ($4,765 in 2009–12, $4,950 in 2013–14, $5,040 in 2014–15, $5,140 in 2015–19); $0.25 \times \$6{,}160$ in 2019–20	1,316 / 1,866
WA	Resource-based formula; LAP allocated mainly by enrollment	0
WI	Achievement-gap aid $2,381 per low-income K–3 pupil (FY2018), spread over K–12	915

Notes: w_{st} is the marginal state aid per FRPL-eligible pupil in FY2024 dollars. Where two values are shown, the first applies to the years in which the formula was suspended or before the rule change. Texas values are for 2015–19 and 2019–20. Sources: EdBuild FundEd "Poverty Funding Policies" and "Base Amount Policies" reports (archived 2016–2019); Texas Education Agency; Michigan House Fiscal Agency; Kansas Legislative Research Department.

Three states did not allocate aid through their formulas in some years. Connecticut distributed legislated block grants and did not use its formula until FY2019. Kansas replaced its formula with block grants in 2015–16 and 2016–17 and adopted a new weighted formula in 2017–18. California replaced revenue limits with the Local Control Funding Formula in 2013–14. Texas raised its basic allotment and replaced the flat 0.2 compensatory weight with tiered weights (0.225–0.275) in 2019–20. I set $w_{st} = 0$ in years without a formula and use the rule in force in each year otherwise.

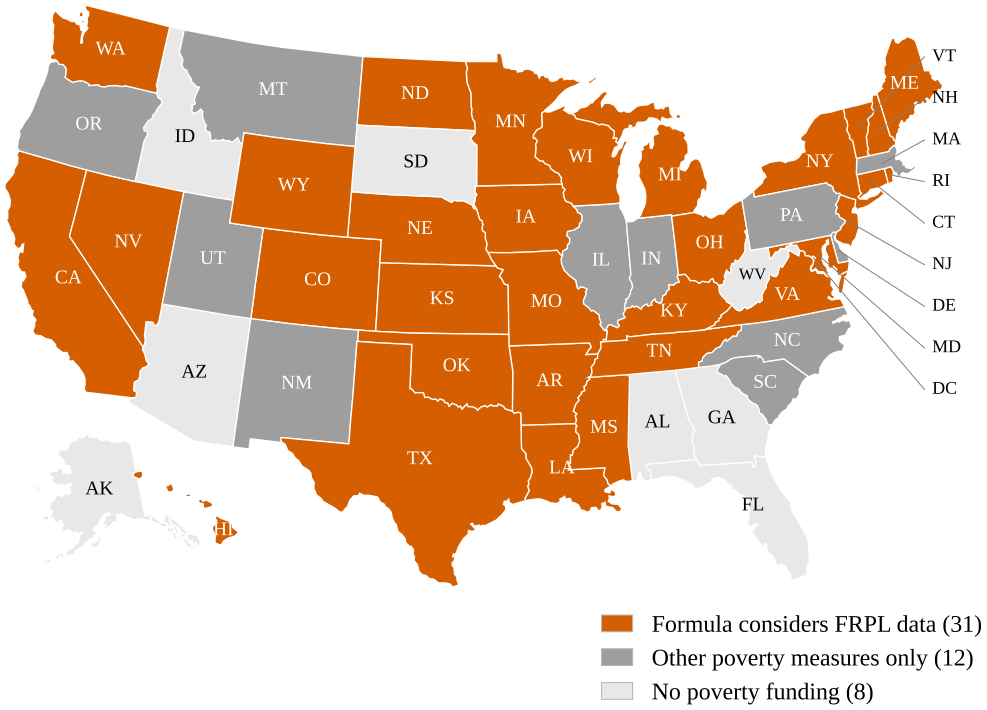
A.2 Appendix Figures and Tables

Figure A.1: Students Eligible for Free or Reduced-Price Meals and Children below 185 Percent of Poverty



Notes: Students eligible for free or reduced-price meals: students in CEP and Provision 2 or 3 schools, plus students certified for free or reduced-price meals in other schools, as a share of public-school enrollment (USDA FNS-742, states with complete reporting). Children below 185 percent of poverty: share of children aged 6–17 with family income below 185 percent of the federal poverty line, the income limit for reduced-price meals (ACS 1-year, table B17024). The dashed vertical line marks the nationwide availability of CEP in 2014–15. School years 2020–21 and 2021–22 (shaded) are omitted because universal-meal waivers make certification counts not comparable; the dashed segment connects 2019–20 and 2022–23. The ACS 1-year estimates for 2020 were not released.

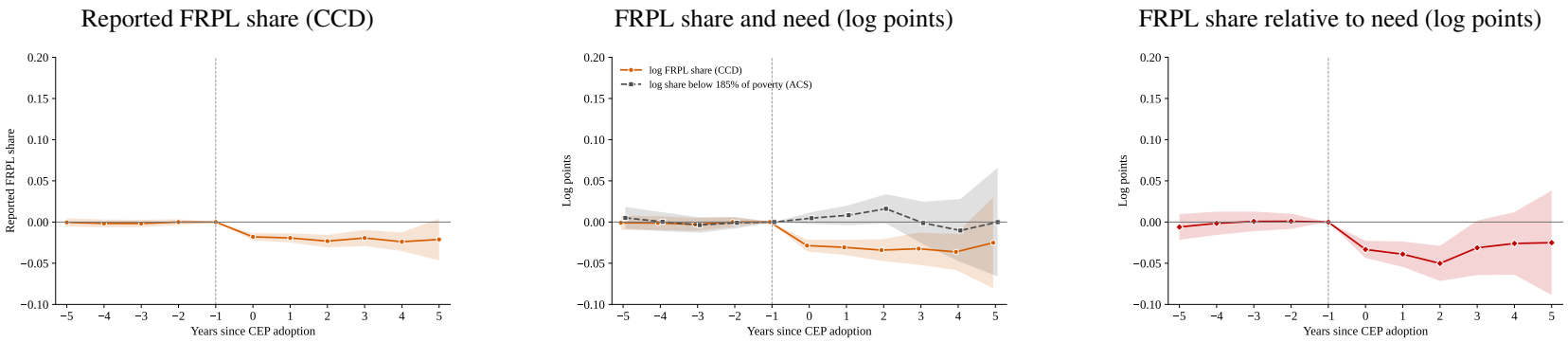
Figure A.2: State Poverty Funding and FRPL Data



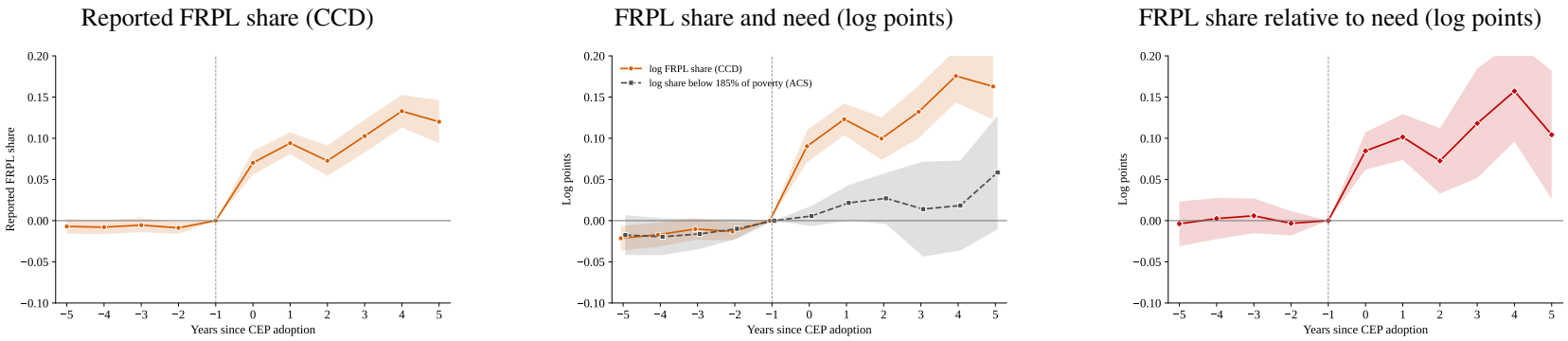
Notes: States and the District of Columbia, classified by how their K–12 funding formulas allocate funding for low-income students. Sources: author's review of state funding formulas; EdBuild FundEd.

Figure A.3: Effects of CEP on Reported FRPL Shares Relative to ACS Need, by CCD Reporting Rule

Panel A. States that report actual FRPL counts (22 states)

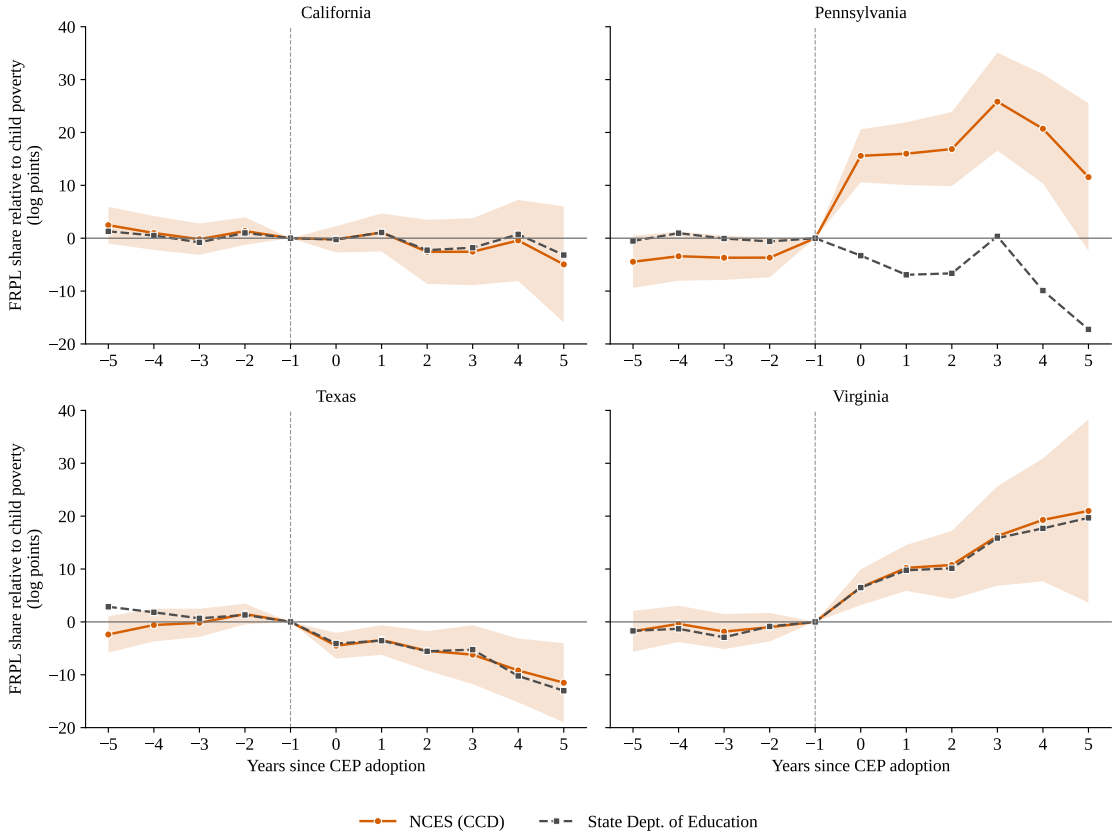


Panel B. States that report imputed FRPL counts (9 states)



Notes: This figure shows event-study estimates from equation (2). Shaded areas are 95 percent confidence intervals, with standard errors clustered by school district. Samples as in Figure 4, restricted to districts with at least 500 children aged 6–17 in every American Community Survey (ACS) window, where ACS estimates are reliable (2,682 districts in Panel A, 699 in Panel B). Need is the ACS share of children aged 6–17 in families below 185 percent of the poverty line (table B17024), the income limit for reduced-price meals; school year t is paired with the ACS 5-year window ending in t. Panel A includes the 22 states that report actual FRPL counts to the Common Core of Data (CCD). Panel B includes the 9 states that report all students in CEP schools, or 1.6 times the identified student percentage, as FRPL eligible. The outcome in the left graph is the FRPL share reported to the CCD. The outcomes in the middle graph are the log FRPL share and the log ACS share of children aged 6–17 with income less than 185 percent of poverty, on the same district-years. The outcome in the right graph is the difference between the two series.

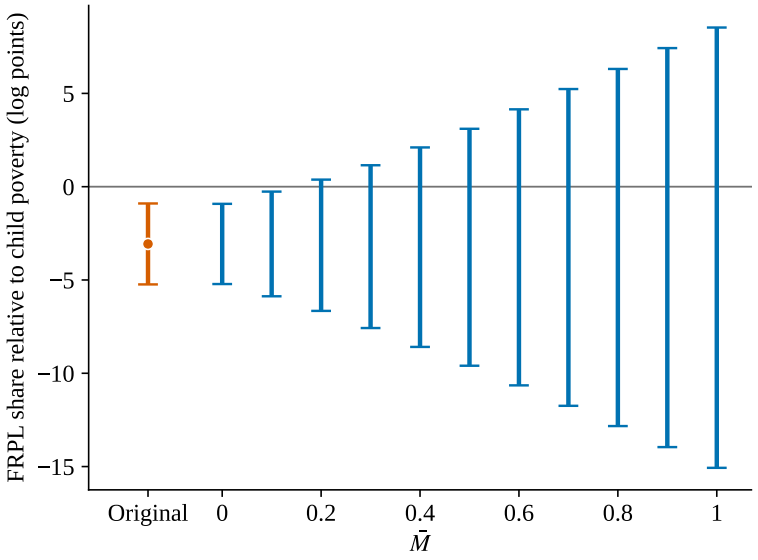
Figure A.4: Checking CCD FRPL Counts against State Sources



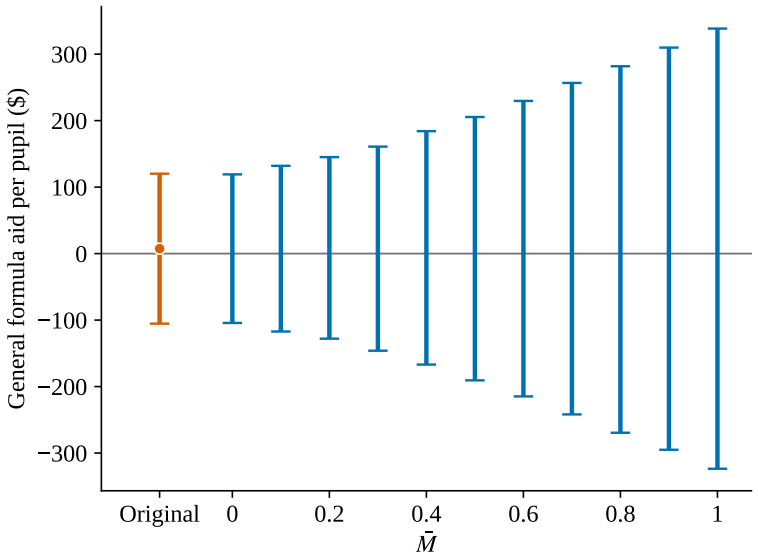
Notes: This figure shows event-study estimates from equation (2) and the associated 95 percent confidence interval. Dashed line: the state department of education's own count relative to child poverty, for the same districts; confidence intervals omitted. State counts: California Department of Education students eligible for free or reduced-price meals, ages 5–17; Pennsylvania Department of Education low-income students; Texas Education Agency economically disadvantaged students; Virginia Department of Education students eligible for free or reduced-price meals.

Figure A.5: Sensitivity of the Main Estimates to Violations of Parallel Trends

Panel A. FRPL share relative to child poverty (log points)

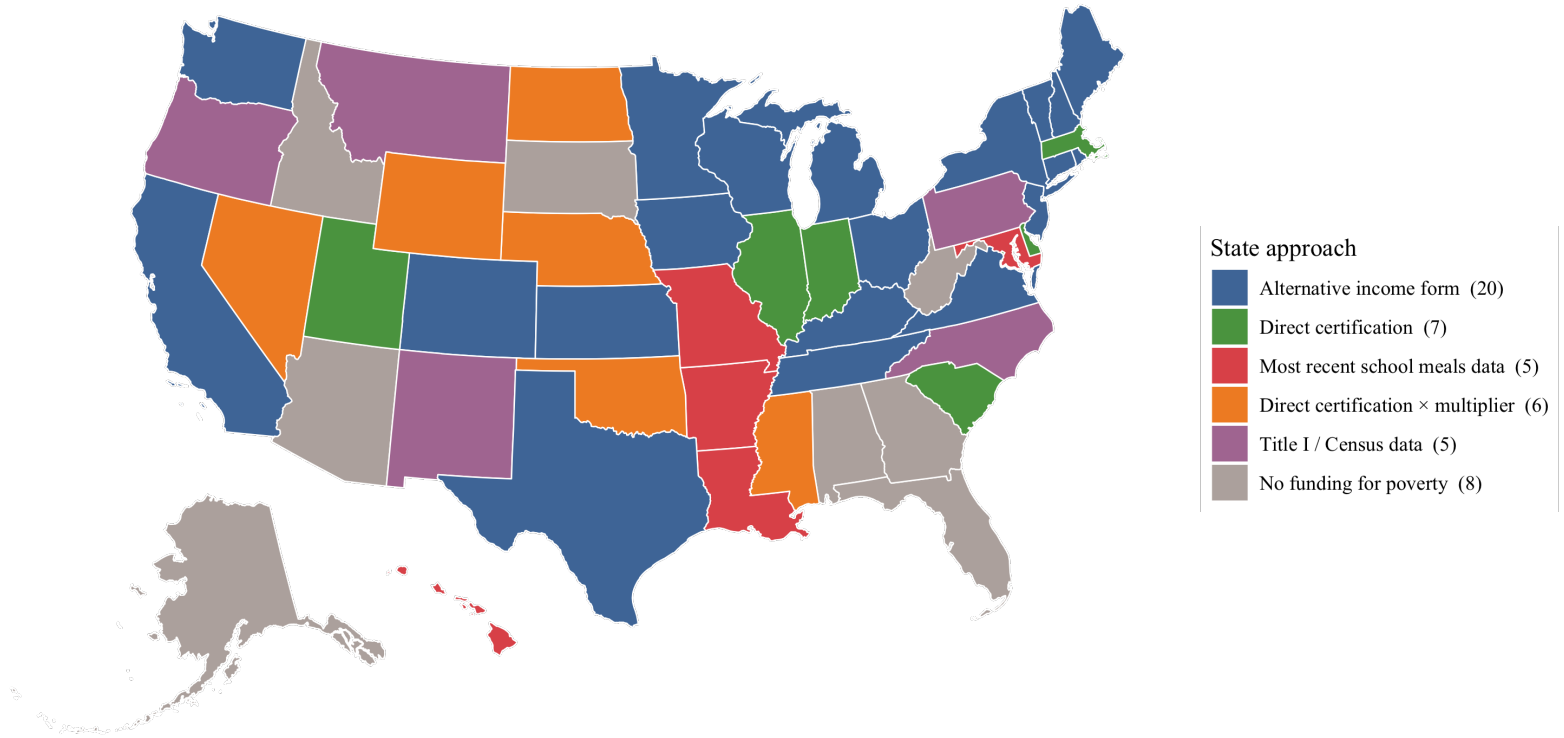


Panel B. General formula aid per pupil ($)



Notes: Robust confidence sets from Rambachan and Roth (2023) for the average effect in years 0–5 after adoption, the post-CEP estimates in Table 3, Panel B, column (4) (Panel A), and Table 4, column (1) (Panel B). Same specification and districts as those tables: states that allocate poverty funding by FRPL and report actual FRPL counts to the CCD. "Original" is the conventional 95 percent confidence interval. For each $\bar{M}$, the bar is the 95 percent robust confidence interval when the post-adoption violation of parallel trends may be up to $\bar{M}$ times the largest change between consecutive pre-adoption years (relative magnitudes). Because state indicators restrict comparisons to districts in the same state, the bounds concern district-level shocks that coincide with adoption. The interval for the FRPL share relative to child poverty excludes zero for $\bar{M} \leq 0.1$ but includes it from $\bar{M} = 0.2$; the interval for general formula aid includes zero at all values of $\bar{M}$.

Figure A.6: Methods That States Use to Handle Changes to Free and Reduced-Price Meal Data



Notes: Author-collected data based on Food Research & Action Center (2017), EdBuild (2020), and Blagg et al. (2025).